The solution is to provide a unique identifier for each participant based on cryptographic algorithms. (You'll learn more about this topic in chapter 5.) All the participants interacting with the blockchain, including a smart contract, have an account number or address that uniquely identifies them:

- Ethereum supports two types of accounts: externally owned accounts (EOAs) and smart contract accounts. Both types of accounts are identified by addresses that are 160 bits, or 40 bytes, long. You can view the account numbers in the Remix IDE.
- Both types of accounts, EOAs and smart contracts, can hold a balance of ethers. Thus, every account has these two *implicit* attributes: `address` and `balance`. You may not find these attributes explicitly declared in your smart contract; `address(this).balance` will get you the balance held in your smart contract.
- An EOA or a smart contract account can invoke a smart contract function by sending a message. This message has two *implied* attributes: `msg.sender` and `msg.value`. Yes, a message can carry a value that is added to the balance held by the smart contract when one of its functions is invoked. The function has to be declared with a payable modifier to be eligible to receive funds.

These concepts are captured in the AccountsDemo.sol smart contract shown in listing 2.2. Copy this smart contract, and deploy it in the Remix IDE. You'll see the screenshot shown in